Artificial Intelligence & Machine Learning – Task 4 Report

Classification Models, Evaluation Metrics & Handling Imbalanced Data

---

1. Introduction

In this task, we moved from regression to **classification**, which is one of the most widely used machine learning techniques in real-world applications such as medical diagnosis, fraud detection, and spam filtering.

The objective of this task was to:

- Build a binary classification model
- Evaluate model performance using appropriate metrics
- Understand why accuracy alone is not sufficient
- Handle class imbalance
- Compare different classification models

2. Dataset Description

For this task, the **Breast Cancer Dataset** from scikit-learn was used. It is a binary classification dataset where:

- **0 → Malignant (Cancerous)**
- **1 → Benign (Non-cancerous)**

The dataset contains multiple numerical features extracted from digitized images of breast masses.

3. Model Implementation

3.1 Logistic Regression

Logistic Regression was used as the baseline model. It is a simple yet powerful linear model for binary classification.

Steps followed:

- Data splitting into training and testing sets
- Feature scaling using StandardScaler
- Model training using Logistic Regression
- Prediction on test data

3.2 Decision Tree Classifier

A Decision Tree model was also implemented to compare performance.

Advantages:

- Easy to interpret
- Captures non-linear relationships

Disadvantages:

- Prone to overfitting

4. Evaluation Metrics

4.1 Confusion Matrix

The confusion matrix provides a detailed breakdown of predictions:

- **True Positive (TP):** Correct positive predictions
- **True Negative (TN):** Correct negative predictions
- **False Positive (FP):** Incorrect positive predictions
- **False Negative (FN):** Missed positive cases

4.2 Precision

Precision measures how many predicted positives are actually correct.

Formula:
Precision = TP / (TP + FP)

4.3 Recall

Recall measures how many actual positives are correctly identified.

Formula:
Recall = TP / (TP + FN)

👉 In medical diagnosis, **Recall is more important** because missing a disease can be dangerous.

4.4 F1 Score

F1 Score is the harmonic mean of Precision and Recall.

Formula:
F1 Score = 2 × (Precision × Recall) / (Precision + Recall)

It is useful when dealing with **imbalanced datasets**.

4.5 Why Accuracy is Misleading

Accuracy alone can give incorrect insights when data is imbalanced.

Example:
If 95% data belongs to one class, the model can achieve 95% accuracy by predicting only that class, which is not useful.

5. ROC Curve & AUC Score

The ROC (Receiver Operating Characteristic) curve shows the trade-off between:

- True Positive Rate (Recall)
- False Positive Rate

The **AUC (Area Under Curve)** represents model performance:

- AUC = 1 → Perfect model
- AUC = 0.5 → Random guessing

A higher AUC indicates a better model.

6. Handling Class Imbalance

Class imbalance is a common problem in real-world datasets.

In this task, imbalance was handled using:

- **Class Weighting (class_weight='balanced')**

This technique assigns more importance to the minority class, improving model performance.

7. Model Comparison

Model	Advantages	Disadvantages
Logistic Regression	Simple, stable, interpretable	Assumes linear relationship
Decision Tree	Easy to understand, handles non-linearity	Can overfit

Observation:

- Logistic Regression provided stable and consistent results
- Decision Tree showed risk of overfitting

8. Final Model Selection

Based on evaluation metrics:

- Logistic Regression performed better in terms of stability and generalization
- It provided a good balance between precision and recall

👉 Therefore, **Logistic Regression was selected as the final model**

9. Conclusion

In this task, we successfully:

- Built classification models
- Evaluated them using proper metrics
- Understood the limitations of accuracy
- Learned the importance of Precision, Recall, and F1-score
- Handled class imbalance effectively
- Compared multiple models and selected the best one

This task provided practical knowledge of real-world machine learning model evaluation and selection.

10. Key Learnings

- Classification is widely used in industry
- Accuracy is not always reliable
- F1-score is important for imbalanced data
- Recall is critical in medical applications
- Model comparison is necessary before final selection